

CMPDI DETAILED PROJECT REPORT (DPR) - Moonidih Underground Longwall

Subsidiary: BCCL | Capacity: 4.5 MT | Geological Basin: Jharia Coalfield, Jharkhand

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This Detailed Project Report (DPR) formulates the mine planning, equipment sizing, and geological stripping layout for Moonidih Underground Longwall under BCCL.

Geological Reserve & Setting: Situated in the Jharia Coalfield, Jharkhand, the mine exploits thick coal seams belonging to the Barakar Formation. Total geological reserves are estimated at 480 Million Tonnes.

Rated Production Capacity: 4.5 MT under mechanized Underground operations. Stripping ratio is projected at Coking Coal Washery, utilizing 42 m3 electric rope shovels, 240T rear dump trucks, and high-capacity bucket wheel systems.

Environmental Mitigation & FMC: Coal evacuation is planned via a First Mile Connectivity (FMC) rapid loading silo with in-pit crushing and a 4.2 km overland conveyor to eliminate surface haul truck emissions.

Technical Parameter	Project Specification	DGMS / Ministry Compliance
Subsidiary / Mine Name	BCCL - Moonidih_Underground	Approved by CIL Board
Rated Annual Capacity	4.5 MT	Environmental Clearance Granted
Mining Method	Underground	High-wall Stability Factor > 1.3
Stripping Ratio	Coking Coal Washery	Continuous Reclamation Mandated
Evacuation Mode	Rapid Loading Silo / FMC	Zero Surface Dust Spillage